

Lesson 12 Weyl group

Monday, April 20, 2026

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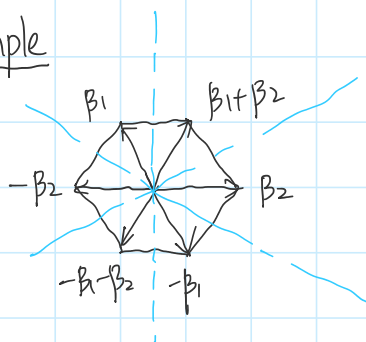
From last time, bases $\xleftrightarrow{H}$ regular decomposition $\Delta = \Delta_+ \cup \Delta_-$
For any bases B , $\exists \gamma \in \mathbb{R}^n$, $(B|\gamma) > 0$.

Def Weyl chamber

The hyperplanes P_α ($\alpha \in \Delta$) partition E into finitely many regions; the connected components of $E \setminus \bigcup_{\alpha \in \Delta} P_\alpha$ are called the (open) Weyl chambers of E . Each regular $\gamma \in E$ therefore belongs to precisely one Weyl chamber.

$\Rightarrow$ Bases $\xleftrightarrow{H}$ Weyl chambers

Example



Def Weyl group

The subgroup of $GL(\mathbb{R}^n)$ generated by all S_α , $\alpha \in \Delta$ is called the Weyl group, denoted by W .

Since $\forall w \in W$, $w(\Delta) = \Delta$, $W \hookrightarrow \text{Sym}(\Delta) \Rightarrow |W| < \infty$.

For any base B , denote $W_B = \langle S_\beta, \beta \in B \rangle$

Lemma

Let B be a base, $\alpha \in B$. Then S_α permutes all $\Delta_+ \setminus \{\alpha\}$.

pf:

Let $\gamma \in \Delta_+$, $\gamma \neq \alpha$. Then $\gamma = k_1\beta_1 + \dots + k_t\beta_t$ for some $\beta_1, \dots, \beta_t \in B$, $k_i > 0$.
 $S_\alpha(\gamma) = k_1\beta_1 + \dots + k_t\beta_t + k_\alpha\alpha$ for some $k_\alpha \in \mathbb{Z}$, i.e. $\exists \beta_i \neq \alpha$ s.t. the coefficients of β_i in γ is the same as in $S_\alpha(\gamma) \Rightarrow S_\alpha(\gamma) \in \Delta_+$.

Lemma

Given any two bases B and B' , $\exists w \in W_B$ s.t. $w(B') = B$.

pf:

Consider $\rho = \frac{1}{2} \sum_{\alpha \in \Delta^+} \alpha \Rightarrow$ For any $\beta \in B$, $S_\beta(\rho) = \rho - \beta$.

Choose γ' s.t. $(\beta' | \gamma') > 0$, we are looking for $w \in W_B$ s.t. $(B | w(\gamma')) > 0$
 $(w(B') | w(\gamma')) > 0$ ← connection.

$\exists w \in W_B$ s.t. $(\rho | w(\gamma'))$ is maximal (W is finite)

Thus $\forall \beta \in B$,

$$(\rho | S_\beta w(\gamma')) \leq (\rho | w(\gamma'))$$

$$\parallel$$

$$(S_\beta \rho | w(\gamma'))$$

$$\parallel$$

$$(\rho - \beta | w(\gamma'))$$

$$\Rightarrow (\beta | w(\gamma')) \geq 0$$

In fact, $(\beta | w(\gamma')) > 0$ since γ' is regular and $w^{-1}(\beta) \in \Delta$

Lemma Any root can be included in some base.

proper subspaces

pf:

Let $\alpha \in \Delta$. Note that $\alpha^\perp \not\subset \bigcup_{\beta \neq \pm \alpha} \beta^\perp$ (otherwise $\alpha^\perp = \bigcup_{\beta \neq \pm \alpha} (\beta^\perp \cap \alpha^\perp)$)

Then we may pick $p \in \alpha^\perp \setminus \bigcup_{\beta \neq \pm \alpha} \beta^\perp$

Choose p' close enough to p s.t. $(p' | \alpha) > 0$, $(p' | \alpha) < \sum_{\beta \neq \pm \alpha} |(p' | \beta)|$

Then α is included in the base corresponding to p'

Corollary $\Delta = W_B B$

Corollary $W = W_B$

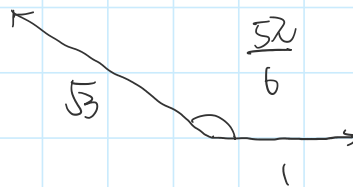
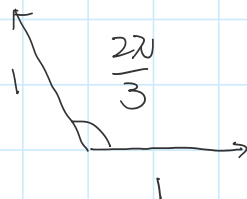
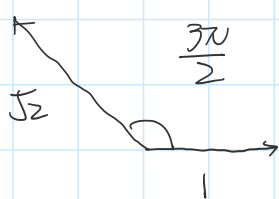
pf:

$\forall \alpha \in \Delta$, $\alpha \in B_i$ for some base B_i . By above, $\exists w \in W_B$ s.t. $w(B_i) = B$.

In particular, $w(\alpha) \in B$, denoted by β . Then as $S_w(\alpha) = w S_\alpha w^{-1}$,

we have $S_\alpha = w^{-1} S_\beta w \in W_B$.

Example irreducible root systems of order 2



Corollary

If α, β are roots, $\beta \neq \pm\alpha$, k is a real number and $\beta + k\alpha$ is a root, then k is an integer.

pf:

By above, $\alpha \in B_1$ for some base $B_1 = \{\alpha_0 = \alpha, \alpha_1, \dots, \alpha_{n-1}\}$. Then as $\beta \in \Delta$, $\beta = \sum_{i=0}^{n-1} k_i \alpha_i$ for some $k_i \in \mathbb{Z}^+$. Then $\beta + k\alpha = (k+k_0)\alpha + \sum_{i=1}^{n-1} k_i \alpha_i$ where each coefficient should be integers $\Rightarrow k \in \mathbb{Z}$.